

# BHAVIN SHAH

[bhavinshah761@gmail.com](mailto:bhavinshah761@gmail.com) | +1- (571)-356-7537 | [Linkedin](#) | [Portfolio](#)

## OBJECTIVE

Versatile Data & AI Engineer bringing 5+ years of end-to-end expertise across distributed ETL pipelines, automated compliance engines, and local/cloud vector-based retrieval systems. Demonstrates deep technical fluency in Python, AWS ecosystem tools, and modern AI orchestration frameworks to deliver scalable, high-availability software solutions.

## EXPERIENCE

### Software Developer

#### SSTech LLC, USA

Jan 2025 – Present

- Architected an event-driven serverless data pipeline using AWS Glue (PySpark), SQS, and Step Functions to ingest, clean, and partition multi-tier financial invoices at scale.
- Automated compliance auditing using AWS Lambda and Amazon Bedrock (Claude Sonnet/Titan) with dual-model failover protection, persisting enriched payloads in S3 Gold and maintaining 100% trace logging in DynamoDB.
- Engineered a serverless automated dialer using AWS Lambda, Amazon Connect, SQS, and DynamoDB to streamline IFS appointment confirmations, implementing SQS queue throttling and mid-call Lambda webhooks for real-time IVR state tracking.
- Designed and deployed SAP BPA automation workflows with strong focus on object oriented principles, reducing manual efforts by 25% across business units.
- Built distributed automation solutions integrating Python APIs, achieving a 40% reduction in manual data handling.
- Implemented robust error-handling and exception recovery, increasing bot execution success rates by 30%.

### Business Technology Analyst

#### Strategic Mindz LLC, USA

Sep 2024 – Jan 2025

- Conducted statistical data analysis using Python and SQL, delivering insights that improved decision-making processes by 25%.
- Built and optimized ETL pipelines for migrating large datasets (Oracle to PostgreSQL), reducing loading time by 50% and improving preprocessing by 45%.
- Delivered scalable solutions to meet business requirements, practicing Agile development methods.

### Product Analyst Intern

#### Clientivity.io, USA

Mar 2024 – Sep 2024

- Processed customer transaction data through Python and SQL, enhancing efficiency and reducing analysis time by 25%.
- Led Agile ceremonies including stand-ups, sprint planning, and Kanban board management for 5+ product features with a 95% completion success rate.
- Effectively gathered and translated technical requirements into actionable reporting solutions.

### Data Engineer I

#### DXC Technology, India

Jun 2019 - Jul 2022

- Designed and implemented scalable data pipelines and distributed systems using Apache Spark and AWS, reducing 1TB+ daily processing time by 40%.
- Automated ETL processes using Python and SQL, enhancing data accuracy by 20%.
- Maintained cloud-based solutions on AWS with optimized cost management, reducing infrastructure spend by 20%.
- Developed user-centric dashboards with Tableau and Power BI, achieving a 95% user satisfaction rate.
- Engaged in object-oriented design and modular system development, ensuring robust and maintainable software solutions.

## SKILLS

- **Languages:** Java, R, SQL, HTML, CSS, Python, JavaScript.
- **Tools & Technologies:** MySQL, SQL server, PostgreSQL, Tableau, Power BI, Microsoft Excel, Oracle, Hadoop, Spark, SAP BPA.
- **Cloud/DevOps:** AWS, GCP, Azure, Databricks, ETL, Airflow, Snowflake, Docker, Kubernetes, Terraform.
- **Libraries:** Pandas, NumPy, Matplotlib, ggplot2, Seaborn, Scikitlearn, PySpark, TensorFlow.
- **AWS services:** S3, Glue, Lambda, SQS, DynamoDB, Amazon Connect, Eventbridge, Step Function, Bedrock.

## EDUCATION

### George Mason University

Fairfax, Virginia, USA

Master's of Science in Information Systems, GPA: 3.87

May 2024

### University of Mumbai

Mumbai, India

Bachelor of Engineering in Electronics Engineering

May 2019

## PROJECTS

---

### **Full-Stack Local RAG Chatbot (FastAPI, LangChain, Streamlit, ChromaDB):**

- Engineered a 100% local, privacy-focused RAG AI chatbot using LangChain, Ollama and Hugging Face embeddings, enabling semantic search and grounded Q&A over local documents with zero data leakage or API costs.
- Architected a decoupled, full stack microservices framework using FastAPI for asynchronous backend orchestration and Streamlit for an interactive chat UI, leveraging ChromaDB for persistent vector retrieval and source citation.

### **Autonomous Web-Scraping RAG Chatbot (Python, LangChain, FastAPI, FAISS, Playwright, BeautifulSoup, Ollama):**

- Engineered an automated web-scraping pipeline using Playwright and BeautifulSoup to extract, clean, and convert dynamic site content into structured markdown, generating vector embeddings stored locally via FAISS.
- Built a robust FastAPI backend integrating LangChain and Ollama (Mistral) to execute context-aware retrieval-augmented generation (RAG) queries, served through a custom chat user interface.

### **AWS S3 Event Triggering System (AWS S3, AWS Lambda, AWS SNS, Python):**

- Developed an automated notification system that triggered real-time email alerts upon file uploads to AWS S3 using Lambda and SNS.
- Improved monitoring speed and operational efficiency by eliminating manual checks and ensuring instant event-driven communication.

### **Covid-19 Testing System Cloud Migration (AWS, GCP, Kubernetes, Terraform, Docker, Cloud SQL):**

- Migrated a legacy healthcare application to cloud using Docker, Kubernetes, and Terraform for automated provisioning and deployment.
- Enhanced application scalability, availability, and response times to support a growing user base in high-demand environments.

### **IPL Data Analysis and ETL Automation (Apache Spark, Databricks, SQL Spark, Python):**

- Built a scalable data pipeline in Spark and Databricks to process IPL datasets and generate real-time performance metrics.
- Automated ETL jobs and scheduled workflows to deliver consistent, high-speed data transformation and reporting.

### **Mobile Enterprise Application Development (Java, Android SDK, SQL, RESTful APIs):**

- Designed and developed a three-tier mobile application with an Android front end and Java-based backend communicating via REST APIs.
- Improved enterprise data access and user experience by implementing secure, real-time client-server communication.

### **Crash Incident Analysis in DMV Area (Python, R, SQL, Machine Learning):**

- Analyzed 25,000+ crash records and used machine learning to identify high-risk locations and peak accident times
- Generated insights that helped inform public safety recommendations for traffic and infrastructure improvements.

### **Telecommunications data (R, Machine Learning, Logistic Regression, Random Forest):**

- Investigated key churn factors including contract type, monthly charges, and demographics, enabling strategy changes that improved customer retention by 10%.
- Implemented logistic regression and random forest models in R, achieving 87% prediction accuracy for customer cancellations.

## ACHIEVEMENTS & CERTIFICATIONS

---

- Awarded with Outstanding Academic Achievement Award from George Mason University in May 2024.
- Completed Google Data Analytics Professional Certificate in Jul 2023.
- Completed AWS Cloud Quest: Cloud Practitioner in Jul 2023
- Awarded with DXC Quarterly Champ Awards in Nov 2020 and then promoted to Data Engineer I role in Oct 2021.

## RESEARCH PAPER

---

- Presented Research paper on 'Design and Implementation of 32-bit RISC processor' in 'National Level Students' Conference on Frontiers in Engineering and Technology Applications – 2019'.